

■ **SENSOR:** Microphone (Web Audio) · Speech recognition · Text-to-speech

The problem

Two under-served groups: people with impaired-but-present speech (post-stroke dysarthria, apraxia, articulation disorders) who need frequent therapy but face an SLP shortage, and people who cannot speak at all who rely on slow or costly assistive tools.

How it works

- Pick a target word/sound; the mic captures each attempt.
- On-device analysis measures pitch, loudness, duration and a clarity score (via speech recognition).
- Live waveform + spectrogram give instant visual feedback.
- The app adapts to the user's own voice and tracks per-sound progress.
- Assistive-voice mode: tap words + predictions → natural TTS speaks for the user.

Example output

Say 'water' — your /r/ was clear, hold the vowel 0.4 s longer

Loudness 62 dB · pitch steady · clarity 78/100 (↑6 this week)

Assistive: mouth a phrase → phone speaks 'I need help, please'

3-day streak · /s/ blends improved 12%

Metrics

Clarity · pitch stability · loudness (dB) · duration · pitch (Hz) · voiced %

Novelty & positioning

Medium–high — combines real-time AI therapy feedback + a personalised assistive voice on-device; AAC and lip-reading exist separately.

Target users

Post-stroke & neuro patients · children with articulation disorders · nonverbal users (autism, ALS, CP)

Assistive / 'cure' feature

Therapy drills (sustained vowel, projection, /r/-/s/-/th/ practice) that measurably improve clarity; AAC voice restores communication.

Research backing

- Smartphone speech therapy improved post-stroke dysarthria (pilot RCT, JMIR 2024)
- Deep-learning dysarthric ASR works best speaker-dependently (Comput. Biol. Med. 2022)
- Real-time acoustic feedback → 2.4× faster gains; SpeechLP (SLP-built)

On-device AI coach

Personal baseline + drift

History & charts

Monthly PDF report

Contact-a-specialist

100% private / offline

Part of the **SenseKit** suite (MotionSense · SpeakSense · HearSense · VisionSense). Screening & self-monitoring only — not a medical diagnosis. Phase 2 = native app with calibrated sensors.